Name: _____

Radioactivity 2 H

Date:

Time:

Total marks available: 47

Total marks achieved: _____

Q1.

Radio waves and gamma radiation are at opposite ends of the electromagnetic spectrum.

Compare how these two electromagnetic radiations are produced.

(6)

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(Total for question = 6 marks)

Q2.

* Nuclear fission and nuclear fusion are two non-renewable sources of energy.

Compare nuclear fission and nuclear fusion as possible sources of energy for generating electricity using a nuclear reactor.

Your comparison should refer to

- the differences between nuclear fission and nuclear fusion
- the relative advantages and difficulties involved in using these sources.

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(Total for question = 6 marks)

Q3.

* Gamma radiation is produced by radioactive decay.

Alpha radiation and beta radiation are also produced by radioactive decay.

Compare the processes of alpha decay and beta decay.

Your answer should include what each radiation is and what effect each decay has on the original nucleus.

(6)

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(Total for question = 6 marks)

Q4.

Radioactive decay can produce gamma radiation and alpha radiation.

Compare the processes of gamma decay and alpha decay.

Your answer should include

- a description of both gamma radiation and alpha radiation
- the effect of each type of decay on the original nucleus.

(6)

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(Total for question = 6 marks)

Q5.

The teacher now investigates the absorption of beta radiation by different thicknesses of aluminium.

The apparatus available is

- a source of beta radiation
- a Geiger-Müller (G-M) tube and counter
- 10 pieces of aluminium, each 0.5 mm thick
- a metre rule.

(i) Sketch a labelled diagram showing the positions of the apparatus when the measurements are being taken.

(2)

(ii) Give the independent variable in this investigation.

(1)

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(iii) Name a quantity that must be kept constant during the investigation.

(1)

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(iv) Strontium-90 is the source of beta minus radiation in this investigation.

Complete the nuclear equation for this emission of beta minus radiation.

(2)

$$^{90}_{38}Sr \longrightarrow \square\ ^{0}\beta + \square\ ^{90}Y$$

(Total for question = 6 marks)

Q6.

Carbon-13 and carbon-14 are isotopes of carbon.

Nuclei of carbon-13 and carbon-14 can be represented by these symbols

$$^{13}_{6}C \quad \text{and} \quad ^{14}_{6}C$$

Complete the table for an atom of carbon–13 and an atom of carbon–14.

(2)

	number of neutrons in the nucleus	number of electrons in orbit around the nucleus
carbon-13		
carbon-14		

(Total for question = 2 marks)

Q7.

Students are given the apparatus shown in Figure 12 and a protractor.

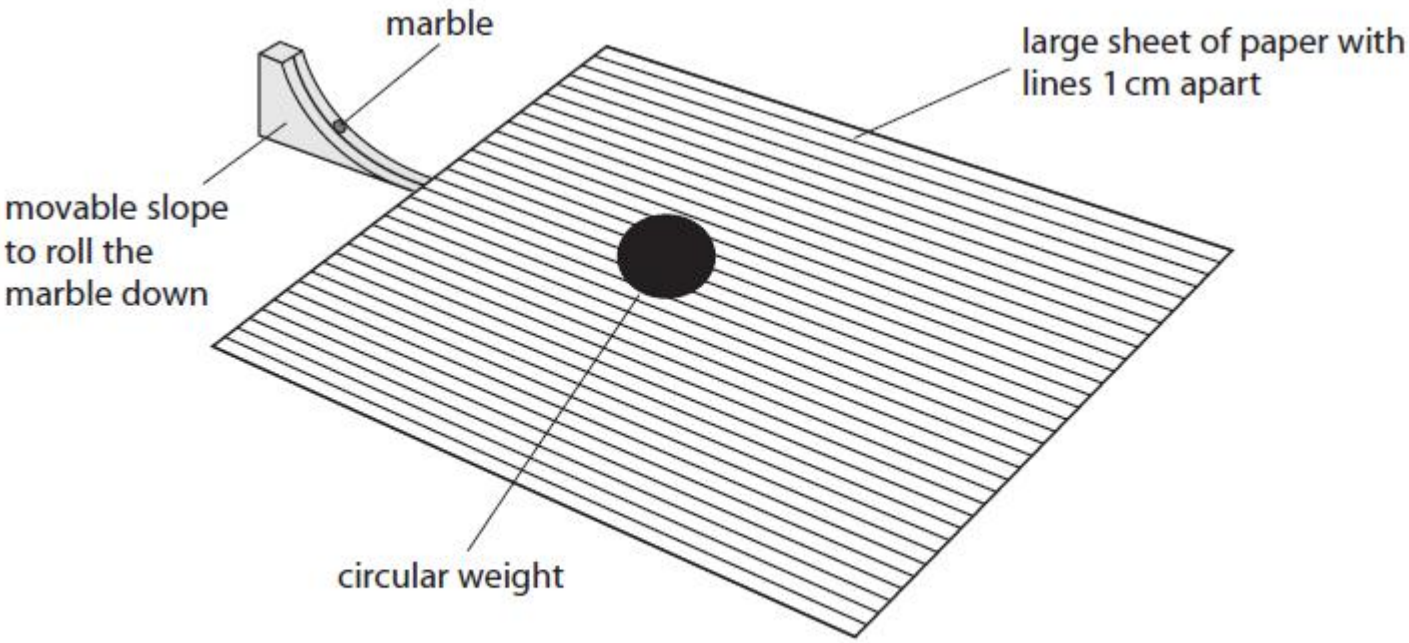

Figure 12

(i) Describe how the students could use the apparatus to model the scattering of alpha particles.

(2)

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(ii) Give **one** limitation of this model.

(1)

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(Total for question = 3 marks)

Q8.

Describe how the energy released in the chain reaction in a nuclear reactor is used to drive a turbine in a nuclear power station.

(3)

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(Total for question = 3 marks)

Q9.

Describe how the thermal energy from nuclear fission can be used to turn the electrical generator in a power station.

(2)

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(Total for question = 2 marks)

Q10.

Stars may originate as a nebula.

(i) Describe the process that then occurs to produce the conditions necessary for nuclear fusion in a new star.

(3)

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(ii) The energy, *E*, released in nuclear fusion is equivalent to loss in mass, *m*, according to the equation.

$$E = mc^2$$

where c is the velocity of light.
$c = 3.00 \times 10^8$ m/s
In 1 second, the energy radiated by the Sun is 3.86×10^{26} J.
Calculate the loss in mass of the Sun in 1 second.

(2)

loss in mass = ... kg

(Total for question = 5 marks)

Q11.

Sometimes food can become contaminated with radioactive substances.

Describe the harmful effects of eating food contaminated with radioactive substances.

(2)

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(Total for question = 2 marks)